Database Schema

Commerce Bank / UMKC Capstone – Fall 2026

MYSQL Database schema for banking application

The schema has six tables: users, categories, payment methods , budgets, transactions, and sessions (contains refresh tokens or cookies for persistence

Users

Stores registered accounts and login credentials.

Column	Type	Constraints	Description
id	CHAR(36)	PK, default UUID()	Unique user identifier
username	VARCHAR(50)	NOT NULL, UNIQUE	Display/login name
email	VARCHAR(255)	NOT NULL, UNIQUE	Login email address
password_hash	VARCHAR(255)	NOT NULL	Hashed Password
created_at	TIMESTAMP	NOT NULL, default now	Account creation time
updated_at	TIMESTAMP	NOT NULL, auto-update	Last profile update time

Categories

Spending/income categories. Rows with user_id = NULL are system defaults visible to every user; a user may also add their own.

Column	Type	Constraints	Description
id	CHAR(36)	PK, default UUID()	Unique category identifier
user_id	CHAR(36)	FK → users.id, NULL allowed	NULL = system default category
name	VARCHAR(100)	NOT NULL	Category label, e.g. “Groceries”
type	VARCHAR(10)	NOT NULL, CHECK income/expense	Whether this category tracks income or spending
created_at	TIMESTAMP	NOT NULL, default now	Row creation time

Payment Methods

Fake/dummy cards used to simulate a payment method on a transaction, per the sponsor requirement to use fabricated card data.

Column	Type	Constraints	Description
id	CHAR(36)	PK, default UUID()	Unique payment method identifier
user_id	CHAR(36)	FK → users.id, NOT NULL	Owning user
nickname	VARCHAR(50)	NOT NULL	Transaction descriptor
card_last4	CHAR(4)	NOT NULL	Last 4 digits only (fabricated)
card_brand	VARCHAR(20)	nullable	e.g. Visa, Mastercard
expiry_date	DATE	NOT NULL	Dummy expiration date
created_at	TIMESTAMP	NOT NULL, default now	Row creation time

Budgets

A spending limit set by a user for a given category and time period.

Column	Type	Constraints	Description
id	CHAR(36)	PK, default UUID()	Unique budget identifier
user_id	CHAR(36)	FK → users.id, NOT NULL	user
category_id	CHAR(36)	FK → categories.id, NOT NULL	Category this budget limits
amount_limit	DECIMAL(12,2)	NOT NULL, CHECK >= 0	Maximum spend allowed for the period
period_type	VARCHAR(10)	NOT NULL, CHECK weekly/monthly	Specific time window (days,weeks,month)
start_date	DATE	NOT NULL	Period start
end_date	DATE	nullable	Period end, if fixed
created_at	TIMESTAMP	NOT NULL, default now	Row creation time

Transactions

The core financial component: each row is one income or expense entry.

Column	Type	Constraints	Description
id	CHAR(36)	PK, default UUID()	Unique transaction identifier
user_id	CHAR(36)	FK → users.id, NOT NULL	User
category_id	CHAR(36)	FK → categories.id, nullable	Type of transaction
payment_method_id	CHAR(36)	FK → payment_methods.id, nullable	Card used
amount	DECIMAL(12,2)	NOT NULL, CHECK > 0	Transaction amount
transaction_type	VARCHAR(10)	NOT NULL, CHECK income/expense	Income/expense type
transaction_date	DATE	NOT NULL, default today	Date the transaction occurred
created_at	TIMESTAMP	NOT NULL, default now	Row insertion time

Sessions (optional -stretch goal)

Column	Type	Constraints	Description
id	CHAR(36)	PK, default UUID()	Unique session identifier
user_id	CHAR(36)	FK → users.id, NOT NULL	Owning user
session_token	VARCHAR(255)	NOT NULL, UNIQUE	token stored in a cookie
expires_at	TIMESTAMP	NOT NULL	Session expiration time
created_at	TIMESTAMP	NOT NULL, default now	Session creation time

AI Live chatbot

LLM API key

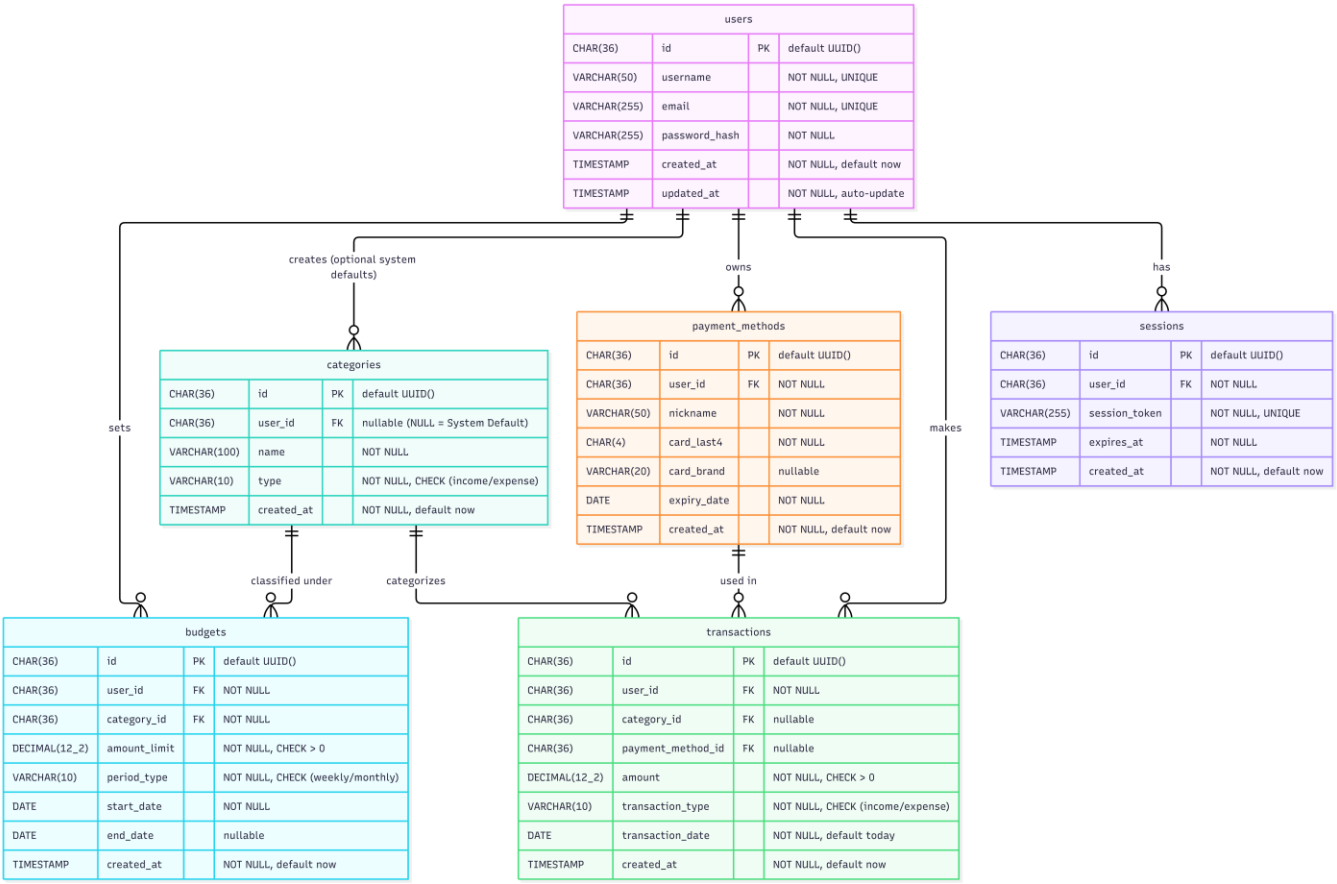
AI Prompts from user

Response from AI

Answer to user who asked question

Column	Type	Constraints	Description
id	CHAR(36)	PK, default UUID()	Unique session identifier
user_id	CHAR(36)	FK → users.id, NOT NULL	Owning user
api_key			
user_prompt			
llm_response			

ER DIAGRAM


users
CHAR(36) id PK default UUID()
VARCHAR(50) username NOT NULL, UNIQUE
VARCHAR(255) email NOT NULL, UNIQUE
VARCHAR(255) password_hash NOT NULL
TIMESTAMP created_at NOT NULL, default now
TIMESTAMP updated_at NOT NULL, auto-update
creates (optional system defaults)
owns
has
sets
makes
categories
CHAR(36) id PK default UUID()
CHAR(36) user_id FK nullable (NULL = System Default)
VARCHAR(100) name NOT NULL
VARCHAR(10) type NOT NULL, CHECK (income/expense)
TIMESTAMP created_at NOT NULL, default now
payment_methods
CHAR(36) id PK default UUID()
CHAR(36) user_id FK NOT NULL
VARCHAR(50) nickname NOT NULL
CHAR(4) card_last4 NOT NULL
VARCHAR(20) card_brand nullable
DATE expiry_date NOT NULL
TIMESTAMP created_at NOT NULL, default now
sessions
CHAR(36) id PK default UUID()
CHAR(36) user_id FK NOT NULL
VARCHAR(255) session_token NOT NULL, UNIQUE
TIMESTAMP expires_at NOT NULL
TIMESTAMP created_at NOT NULL, default now
classified under
categorizes
used in
budgets
CHAR(36) id PK default UUID()
CHAR(36) user_id FK NOT NULL
CHAR(36) category_id FK NOT NULL
DECIMAL(12_2) amount_limit NOT NULL, CHECK > 0
VARCHAR(10) period_type NOT NULL, CHECK (weekly/monthly)
DATE start_date NOT NULL
DATE end_date nullable
TIMESTAMP created_at NOT NULL, default now
transactions
CHAR(36) id PK default UUID()
CHAR(36) user_id FK NOT NULL
CHAR(36) category_id FK nullable
CHAR(36) payment_method_id FK nullable
DECIMAL(12_2) amount NOT NULL, CHECK > 0
VARCHAR(10) transaction_type NOT NULL, CHECK (income/expense)
DATE transaction_date NOT NULL, default today
TIMESTAMP created_at NOT NULL, default now